

AI / ML / Generative AI Problem Statements

Title	Topic	Architecture
Student Performance Prediction	Classification	Data → Preprocessing → Feature Engineering → Classification Model → Prediction
Employee Attrition Prediction	Classification	HR Data → Preprocessing → Feature Selection → Random Forest → Attrition Prediction
Loan Approval Prediction	Classification	Applicant Data → Preprocessing → Classification Model → Approved/Rejected
Email Spam Detection	Classification	Email Text → TF-IDF → Classifier → Spam/Ham
Customer Churn Prediction	Classification	Customer Data → Preprocessing → Classification Model → Churn Prediction
House Price Prediction	Regression	Property Data → Preprocessing → Feature Engineering → Regression Model → Price Prediction
Sales Revenue Prediction	Regression	Sales Data → Preprocessing → Feature Engineering → Regression Model → Revenue Forecast
Used Car Price Prediction	Regression	Car Data → Preprocessing → Regression Model → Price Prediction
Electricity Consumption Prediction	Regression	Historical Usage → Feature Engineering → Regression Model → Consumption Prediction
Crop Yield Prediction	Regression	Agricultural Data → Preprocessing → Regression Model → Yield Prediction
Student Success Prediction using ANN	ANN	Input Features → Dense Layers → ReLU → Output Layer → Prediction
Customer Churn Prediction using ANN	ANN	Customer Features → Dense Layers → ReLU → Sigmoid → Churn Class
Fraud Detection using ANN	ANN	Transaction Data → Dense Layers → Activation → Output → Fraud/Legitimate
Energy Consumption Prediction using ANN	ANN	Energy Data → Normalization → Dense Network → Regression Output
Disease Risk Prediction using ANN	ANN	Patient Features → Dense Layers → ReLU → Softmax → Risk Class
Handwritten Digit Recognition	CNN	Image → Convolution → ReLU → Pooling → Flatten → Dense → Softmax
Plant Disease Detection	CNN	Leaf Image → Conv2D → ReLU → Pooling → Dense → Disease Class
Waste Classification	CNN	Waste Image → Convolution → Pooling → Flatten → Dense → Waste Class
Traffic Sign Recognition	CNN	Traffic Image → CNN Feature Extraction → Dense → Sign Class
Fruit Quality Classification	CNN	Fruit Image → Convolution → Pooling → Dense → Fresh/Damaged/Rotten
Synthetic Handwritten Digit Generation	GAN	Noise → Generator → Fake Image; Real/Fake → Discriminator → Adversarial Training
Synthetic Face Generation	GAN	Latent Noise → Generator → Face Image; Image → Discriminator → Real/Fake
Synthetic Fashion Image Generation	GAN	Noise → Generator → Fashion Image → Discriminator → GAN Training
GAN-Based Data Augmentation	GAN	Real Images → GAN Training → Synthetic Images → Augmented Dataset
Image-to-Image Translation	GAN	Source Image → Generator → Target Image → Discriminator
AI Resume Analyzer	Prompt Engineering	Resume → Structured Prompt → LLM → Skills/Experience/Job Recommendations
AI Interview Question Generator	Prompt Engineering	Resume + Job Role → Few-Shot Prompt → LLM → Interview Questions
AI Code Review Assistant	Prompt Engineering	Source Code → Role-Based Prompt → LLM → Bugs/Issues/Recommendations

Title	Topic	Architecture
SQL Query Generator	Prompt Engineering	Natural Language → Prompt → LLM → SQL Query → Database
Meeting Summarization Assistant	Prompt Engineering	Transcript → Structured Prompt → LLM → Summary/Decisions/Actions
AI Study Assistant	Generative AI	Student Query → LLM → Explanation → Examples → Quiz/Study Plan
AI Resume Builder	Generative AI	User Details → LLM → Prompt Template → Professional Resume
AI Question Paper Generator	Generative AI	Subject + Syllabus → Prompt → LLM → Questions → Difficulty/Marks
AI Report Generator	Generative AI	Structured Data → Analysis → LLM → Business Report
Multimodal AI Assistant	Generative AI	Text/Image → Vision-Language Model → Reasoning → Generated Response
Product Review Sentiment Analysis	Sentiment Analysis	Review Text → Cleaning → TF-IDF/Embeddings → Classifier → Sentiment
Restaurant Review Analyzer	Sentiment Analysis	Reviews → NLP Preprocessing → Sentiment Model → Positive/Negative/Neutral
Student Feedback Analysis	Sentiment Analysis	Feedback → NLP → Sentiment Classification → Insights
Aspect-Based Sentiment Analysis	Sentiment Analysis	Review → Aspect Extraction → Sentiment Classification → Aspect-Level Results
Brand Sentiment Analysis	Sentiment Analysis	Social Text → NLP → Sentiment Model → Brand Sentiment Dashboard
College Information Assistant	RAG	Documents → Chunking → Embeddings → Vector DB → Retrieval → LLM → Answer
HR Policy RAG Assistant	RAG	HR PDFs → Chunking → Embeddings → FAISS/Chroma → Retrieval → LLM
Research Paper Assistant	RAG	Research PDFs → Chunking → Embeddings → Vector Store → Retrieval → LLM
Technical Documentation Assistant	RAG	Documentation → Chunking → Embeddings → Vector DB → Retriever → LLM
Product Manual Chatbot	RAG	Manuals → Text Extraction → Embeddings → Vector DB → Retrieval → LLM
Plant Disease Advisory System	CNN + RAG + Generative AI	Leaf Image → CNN → Disease Class → RAG → LLM → Advisory
AI Resume Career Assistant	NLP + Classification + RAG + GenAI	Resume → NLP → Job Classification → RAG → LLM → Career Recommendations
Intelligent Customer Feedback Analytics	Sentiment + Classification + RAG + GenAI	Reviews → Sentiment → Classification → Retrieval → LLM → Response
Intelligent College Assistant	RAG + Prompt Engineering + GenAI	College Documents → Vector DB → Retrieval → Prompt → LLM → Grounded Answer
Synthetic Data Intelligence System	GAN + Classification + Regression	Real Data → GAN → Synthetic Data → ML Models → Performance Comparison